THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	$F = 2 + E - V = 8$
2	**(c)** 4	Each segment becomes 4: $4^1 = 4$
3	**(b)** $\frac{9}{16}$	Each stage keeps $\frac{3}{4}$: $(3/4)^2$
4	**(a)** 6	$6(n-2)^2 = 6 \times 1$
5	**(a)** two circles and one rectangle	Unrolling the curved surface gives a rectangle
6	**(b)** circle (with a dot at the centre)	Looking down, the base circle with the apex at centre
7	**(a)** 7	$F = 2 + E - V = 7$
8	**(d)** 16	Each segment becomes 4: $4^2 = 16$
9	**(a)** $\frac{27}{64}$	Each stage keeps $\frac{3}{4}$: $(3/4)^3$
10	**(a)** 24	$6(n-2)^2 = 6 \times 4$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Edges $= 3n = 21$ so $n = 7$ (1); faces $= n + 2 = 9$ (1).	1+1
12	$F = 10$, $V = 10$, $E = 18$ (1); $F + V - E = 10 + 10 - 18 = 2$ (1).	1+1
13	$V = 2 + E - F$ (1) $= 14$ (1).	1+1
14	Edges $= 3n = 27$ so $n = 9$ (1); faces $= n + 2 = 11$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $12(n-2) = 12 \times 1 = 12$ (1+1); (ii) $6(n-2)^2 = 6 \times 1 = 6$ (1).	2+1
16	Segments $= 4^3 = 64$ (1); each $= L/3^3 = 1$ cm (1); total $= 64$ cm (1).	1+1+1
17	Shaded area $= 64 \times (3/4)^2 = 36$ cm² (2); removed $= 28$ cm² (1).	2+1
	OR: (i) $12(n-2) = 12 \times 2 = 24$ (1+1); (ii) $6(n-2)^2 = 6 \times 4 = 24$ (1).	2+1
18	Segments $= 4^4 = 256$ (1); each $= L/3^4 = 1$ cm (1); total $= 256$ cm (1).	1+1+1
	OR: Shaded area $= 256 \times (3/4)^2 = 144$ cm² (2); removed $= 112$ cm² (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Prism: $F = 5$, $V = 6$, $E = 9$ (2); $F + V - E = 5 + 6 - 9 = 2$ (1); pyramid: $F = 4$, $V = 4$, $E = 6$ (1); $F + V - E = 4 + 4 - 6 = 2$ (1).	2+1+1+1
	OR: Three faces: 8 (1); two faces: $12(9-2) = 84$ (1); one face: $6(9-2)^2 = 294$ (1); none: $(9-2)^3 = 343$ (1); total $= 8 + 84 + 294 + 343 = 729$ (1).	1+1+1+1+1
20	Prism: $F = 6$, $V = 8$, $E = 12$ (2); $F + V - E = 6 + 8 - 12 = 2$ (1); pyramid: $F = 5$, $V = 5$, $E = 8$ (1); $F + V - E = 5 + 5 - 8 = 2$ (1).	2+1+1+1
	OR: Three faces: 8 (1); two faces: $12(10-2) = 96$ (1); one face: $6(10-2)^2 = 384$ (1); none: $(10-2)^3 = 512$ (1); total $= 8 + 96 + 384 + 512 = 1000$ (1).	1+1+1+1+1

Mistakes to avoid:

- Counting faces, edges and vertices of a prism and pyramid with each other's formulas (a pyramid has $2n$ edges, a prism $3n$).
- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n - 2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** one circle and one sector	Curved surface opens into a sector
2	**(d)** rectangle	Side view of upright cylinder
3	**(a)** 5	$F = 2 + E - V = 5$
4	**(c)** 64	Each segment becomes 4: $4^3 = 64$
5	**(d)** $\frac{81}{256}$	Each stage keeps $\frac{3}{4}$: $(3/4)^4$
6	**(d)** 54	$6(n-2)^2 = 6 \times 9$
7	**(b)** two triangles and three rectangles	Two bases and three lateral faces
8	**(b)** triangle	Isosceles triangle
9	**(b)** 7	$F = 2 + E - V = 7$
10	**(a)** 256	Each segment becomes 4: $4^4 = 256$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$F = 11$, $V = 11$, $E = 20$ (1); $F + V - E = 11 + 11 - 20 = 2$ (1).	1+1
12	$V = 2 + E - F$ (1) $= 16$ (1).	1+1
13	Edges $= 3n = 33$ so $n = 11$ (1); faces $= n + 2 = 13$ (1).	1+1
14	$F = 13$, $V = 13$, $E = 24$ (1); $F + V - E = 13 + 13 - 24 = 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $12(n-2) = 12 \times 3 = 36$ (1+1); (ii) $6(n-2)^2 = 6 \times 9 = 54$ (1).	2+1
16	Segments $= 4^2 = 16$ (1); each $= L/3^2 = 1$ cm (1); total $= 16$ cm (1).	1+1+1
17	Shaded area $= 256 \times (3/4)^3 = 108$ cm² (2); removed $= 148$ cm² (1).	2+1
	OR: (i) $12(n-2) = 12 \times 4 = 48$ (1+1); (ii) $6(n-2)^2 = 6 \times 16 = 96$ (1).	2+1
18	Segments $= 4^2 = 16$ (1); each $= L/3^2 = 2$ cm (1); total $= 32$ cm (1).	1+1+1
	OR: Shaded area $= 512 \times (3/4)^3 = 216$ cm² (2); removed $= 296$ cm² (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Prism: $F = 7$, $V = 10$, $E = 15$ (2); $F + V - E = 7 + 10 - 15 = 2$ (1); pyramid: $F = 6$, $V = 6$, $E = 10$ (1); $F + V - E = 6 + 6 - 10 = 2$ (1).	2+1+1+1
	OR: Three faces: 8 (1); two faces: $12(12-2) = 120$ (1); one face: $6(12-2)^2 = 600$ (1); none: $(12-2)^3 = 1000$ (1); total $= 8 + 120 + 600 + 1000 = 1728$ (1).	1+1+1+1+1
20	Prism: $F = 8$, $V = 12$, $E = 18$ (2); $F + V - E = 8 + 12 - 18 = 2$ (1); pyramid: $F = 7$, $V = 7$, $E = 12$ (1); $F + V - E = 7 + 7 - 12 = 2$ (1).	2+1+1+1
	OR: Three faces: 8 (1); two faces: $12(15-2) = 156$ (1); one face: $6(15-2)^2 = 1014$ (1); none: $(15-2)^3 = 2197$ (1); total $= 8 + 156 + 1014 + 2197 = 3375$ (1).	1+1+1+1+1

Mistakes to avoid:

- Counting faces, edges and vertices of a prism and pyramid with each other's formulas (a pyramid has $2n$ edges, a prism $3n$).
- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n - 2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{243}{1024}$	Each stage keeps $\frac{3}{4}$: $(3/4)^5$
2	**(a)** 96	$6(n-2)^2 = 6 \times 16$
3	**(c)** one square and four triangles	Base and four slant faces
4	**(d)** square	Looking down at the top face
5	**(b)** 14	$F = 2 + E - V = 14$
6	**(c)** 1024	Each segment becomes 4: $4^5 = 1024$
7	**(a)** 150	$6(n-2)^2 = 6 \times 25$
8	**(b)** four equilateral triangles	Four triangular faces
9	**(c)** square	Same as the face
10	**(a)** 32	$F = 2 + E - V = 32$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$V = 2 + E - F$ (1) $= 20$ (1).	1+1
12	Edges $= 3n = 42$ so $n = 14$ (1); faces $= n + 2 = 16$ (1).	1+1
13	$F = 15$, $V = 15$, $E = 28$ (1); $F + V - E = 15 + 15 - 28 = 2$ (1).	1+1
14	$V = 2 + E - F$ (1) $= 24$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $12(n-2) = 12 \times 5 = 60$ (1+1); (ii) $6(n-2)^2 = 6 \times 25 = 150$ (1).	2+1
16	Segments $= 4^3 = 64$ (1); each $= L/3^3 = 2$ cm (1); total $= 128$ cm (1).	1+1+1
17	Shaded area $= 1024 \times (3/4)^3 = 432$ cm² (2); removed $= 592$ cm² (1).	2+1
	OR: (i) $12(n-2) = 12 \times 6 = 72$ (1+1); (ii) $6(n-2)^2 = 6 \times 36 = 216$ (1).	2+1
18	Segments $= 4^4 = 256$ (1); each $= L/3^4 = 2$ cm (1); total $= 512$ cm (1).	1+1+1
	OR: Shaded area $= 1024 \times (3/4)^4 = 324$ cm² (2); removed $= 700$ cm² (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Prism: $F = 9$, $V = 14$, $E = 21$ (2); $F + V - E = 9 + 14 - 21 = 2$ (1); pyramid: $F = 8$, $V = 8$, $E = 14$ (1); $F + V - E = 8 + 8 - 14 = 2$ (1).	2+1+1+1
	OR: Three faces: 8 (1); two faces: $12(20-2) = 216$ (1); one face: $6(20-2)^2 = 1944$ (1); none: $(20-2)^3 = 5832$ (1); total $= 8 + 216 + 1944 + 5832 = 8000$ (1).	1+1+1+1+1
20	Prism: $F = 10$, $V = 16$, $E = 24$ (2); $F + V - E = 10 + 16 - 24 = 2$ (1); pyramid: $F = 9$, $V = 9$, $E = 16$ (1); $F + V - E = 9 + 9 - 16 = 2$ (1).	2+1+1+1
	OR: Three faces: 8 (1); two faces: $12(25-2) = 276$ (1); one face: $6(25-2)^2 = 3174$ (1); none: $(25-2)^3 = 12167$ (1); total $= 8 + 276 + 3174 + 12167 = 15625$ (1).	1+1+1+1+1

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